SENG 201
Software Engineering Foundations

[Week 2, Monday]



What did we see last week?

- Purpose of software engineering
- Development processes and activities
- Software development lifecycle (SDLC)
- Requirements, design, testing, deployment, maintenance
- UML

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A first Java class



- → Hello, World!
- → `main` method
- → `System.out.println`
- → Documentation

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Hello, World!

```java
class HelloWorld {

    void main() {
        // a comment!
        System.out.println("Hello, World!");

        int courseNumber = 201;
        String courseDomain = "SENG";
        String message = "Welcome to " + courseDomain + courseNumber;
        System.out.println(message);
    }

}
```


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CostCalculator

```
/*
 * Warning:
 * In the real world,
 * NEVER use floating-point arithmetic
 * to compute prices.
 */

class CostCalculator {

    int discountThreshold = 100;
    double discount = 0.10;
    double gstRate = 0.15;

    void main() {
        showPurchaseInformation("Watermelons (for math class)", 30, 150, true);
        showPurchaseInformation("Laptop", 2500, 1, false);

        String[] penColors = { "blue", "black", "red", "green", "pink" };
        for (String penColor : penColors) {
            showPurchaseInformation(penColor + " pens", 1.5, 200, true);
        }
    }

    double computePrice(int nbItems, double unitPrice) {
```


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Documentation

Line comments (to the right or just above)

```
// this is a line comment
```

```
// comment about x
int x;
int y; // comment about y
```

Block comments (multiple lines or inside a line)

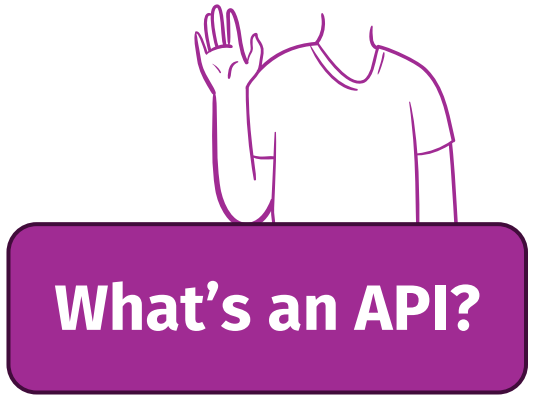
```
/* this is a block comment */
/*
   also a
   block comment
 */
```

```
/*
 * Standalone comment
 * E.g., code license
 */

void omitted() { /* ... */ }
```


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API Reference (Javadoc)



Code comment above API to document using structured tags

```
/** Javadoc */
```

description

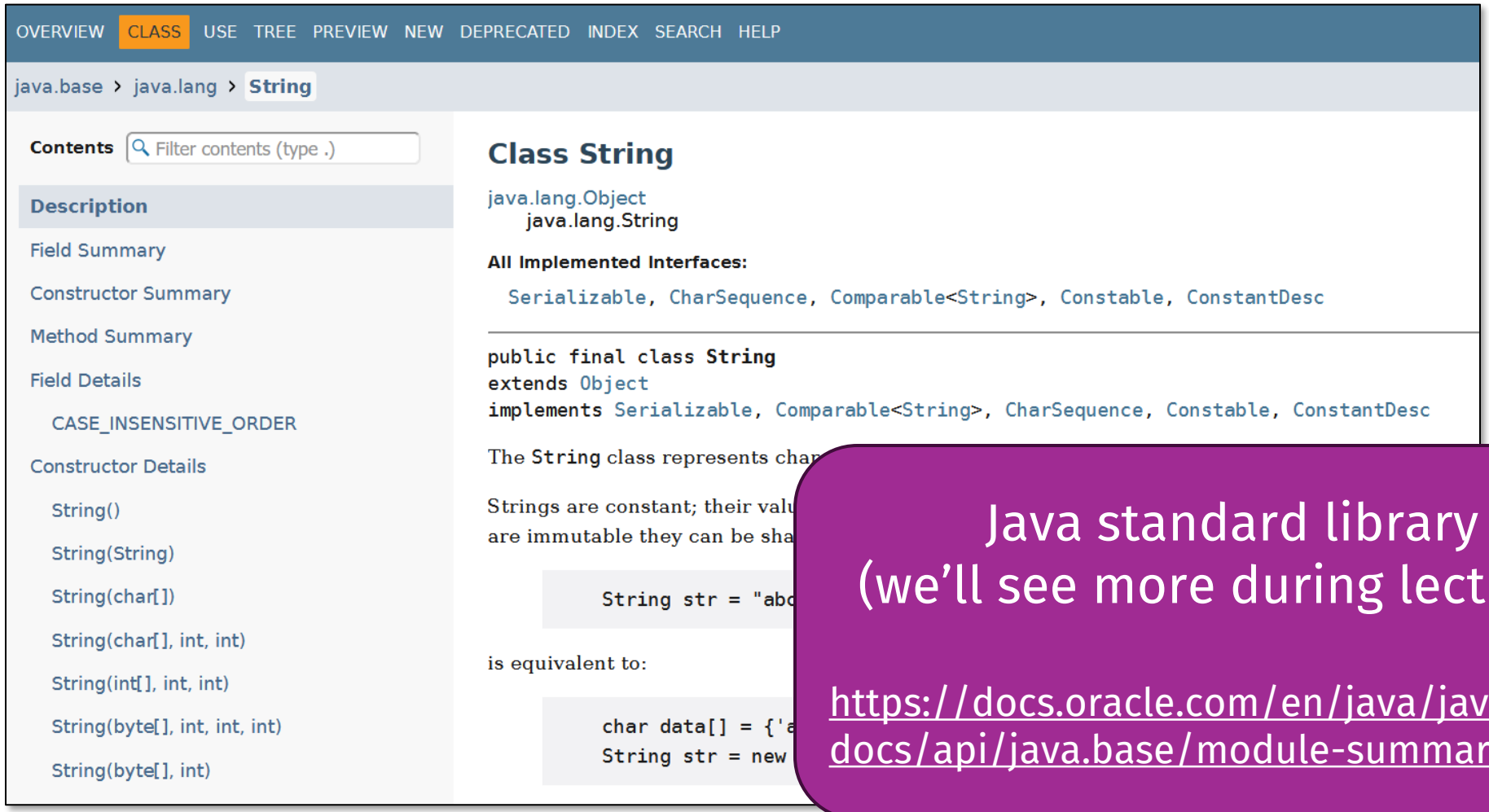
parameters

return value

```
/**
 * Computes the total price, including a potential discount.
 *
 * @param nbItems number of items
 * @param unitPrice price in NZD of each item
 * @return the total price in NZD
 */
double computePrice(int nbItems, double unitPrice) {
    if (nbItems <= 0 || unitPrice < 0) {
```


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API Reference (Javadoc)

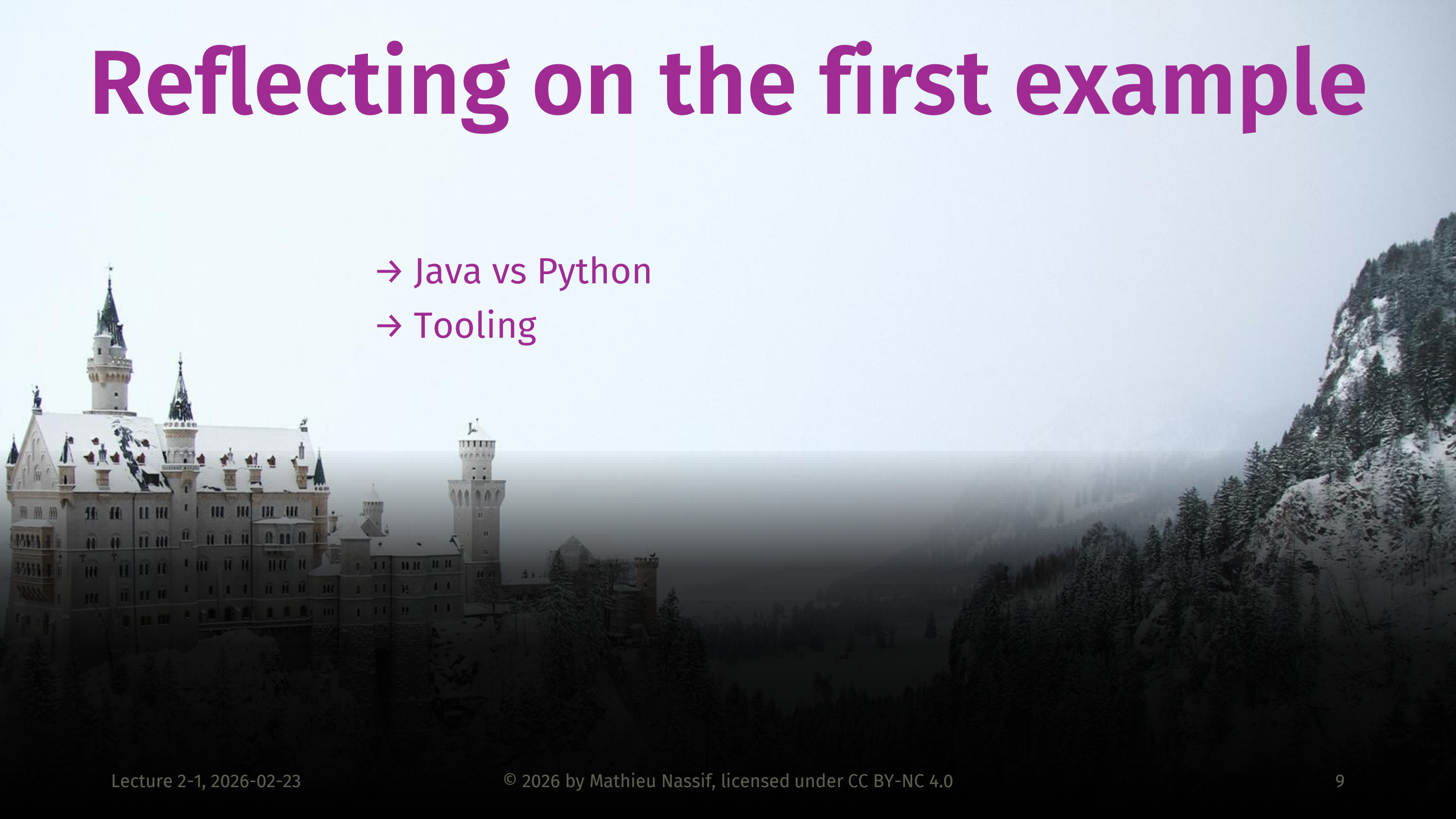


Java standard library
(we'll see more during lectures)

https://docs.oracle.com/en/java/javase/25/docs/api/java.base/module-summary.html

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Reflecting on the first example



→ Java vs Python

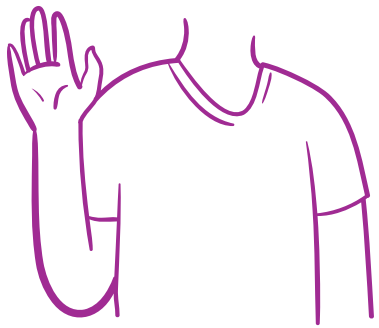
→ Tooling

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Java and Python

Differences

- Code must be in a class*
- Entry point: `main`
- Variable declarations
- Explicit types
- Required ; and {}
- Optional indentation/whitespace
- Compiled language



Similarities

- Started in the mid-1990s
- Object-oriented languages
- Popular and mature
- Similar features
  - ✓ classes, functions
  - ✓ for, while, if, …
  - ✓ exceptions

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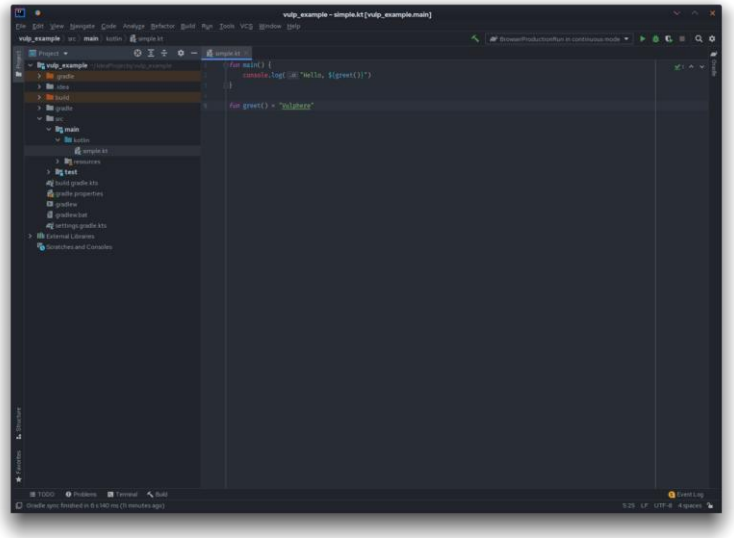
What you need for Java

1. Java editor
2. Java compiler
3. Java virtual machine (JVM)

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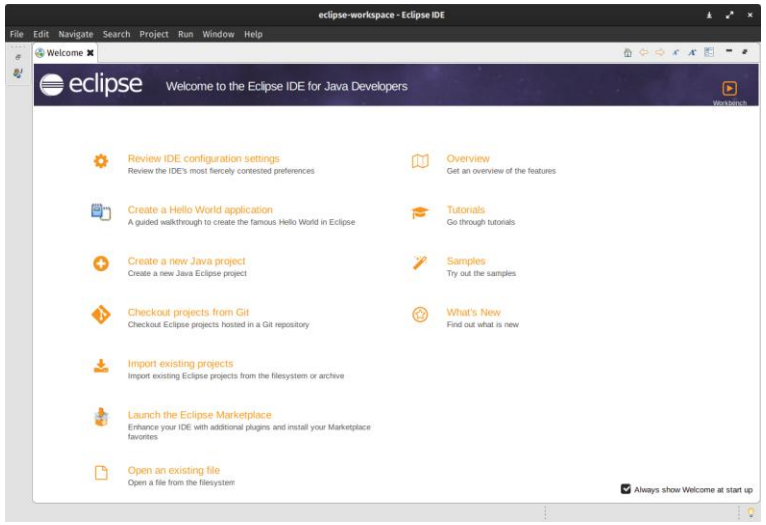
Writing Java programs

Free license (education)



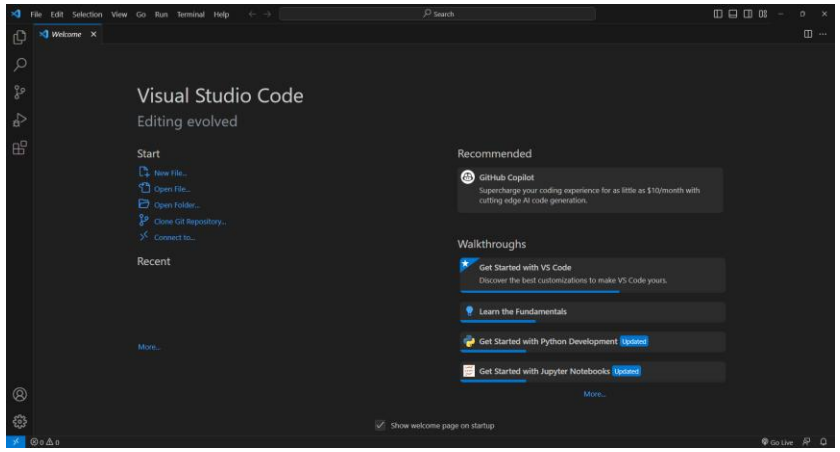
IntelliJ IDEA
JETBRAINS IDE

Free, open source



eclipse

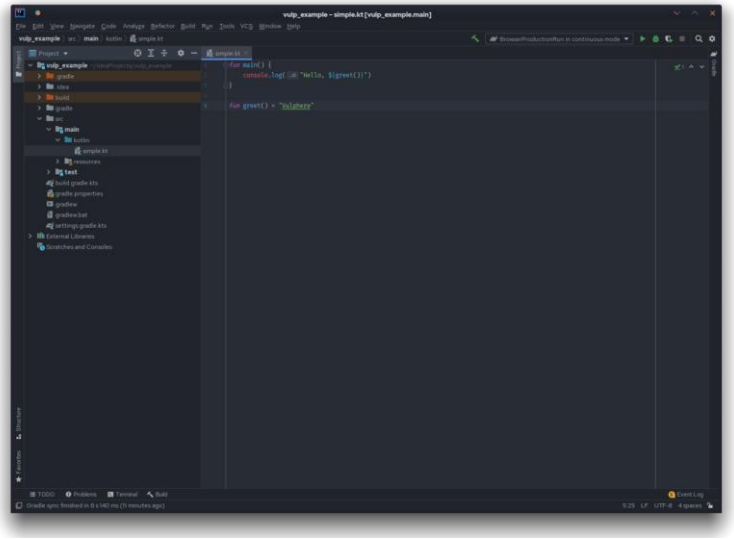
Free (based on OSS)



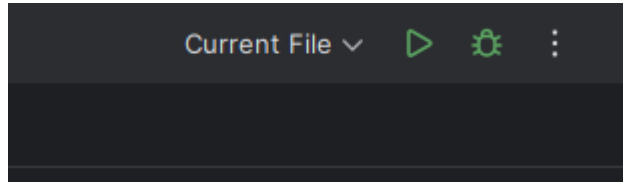
Visual Studio Code

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Running Java programs (IDE)

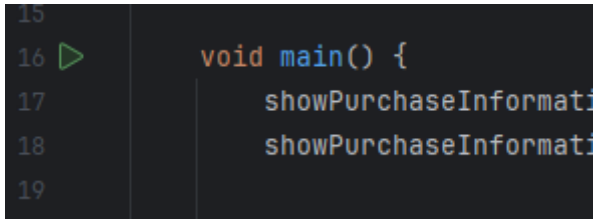


IntelliJ IDEA
JETBRAINS IDE



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Running Java programs (console)

1. Download and install JDK (compiler + runtime)
   - Latest LTS: Java 25
   - Adoptium: https://adoptium.net/temurin/releases/



2. Compile .java files

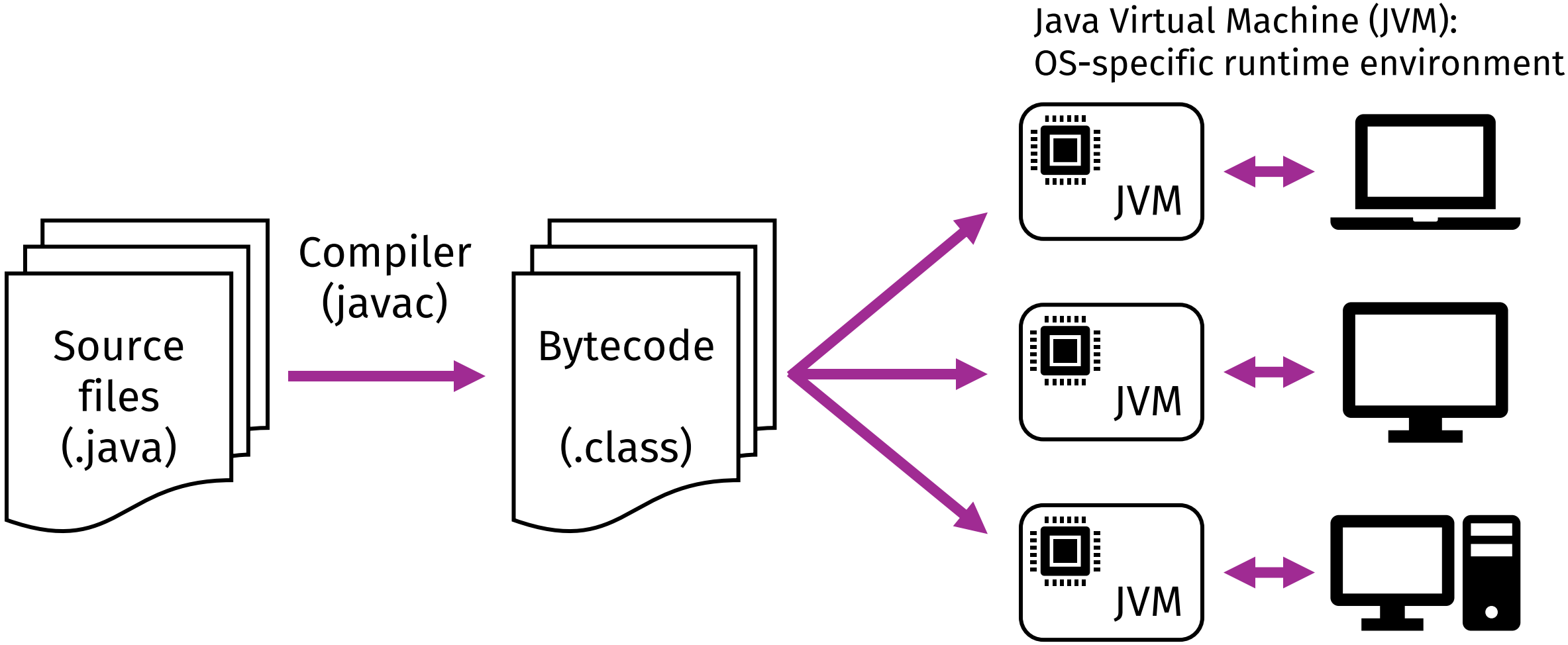
```
javac -cp src/ -d out/ src/lecture2_1/HelloWorld.java
```

3. Run .class files

```
java -cp out/ lecture2_1.HelloWorld
```

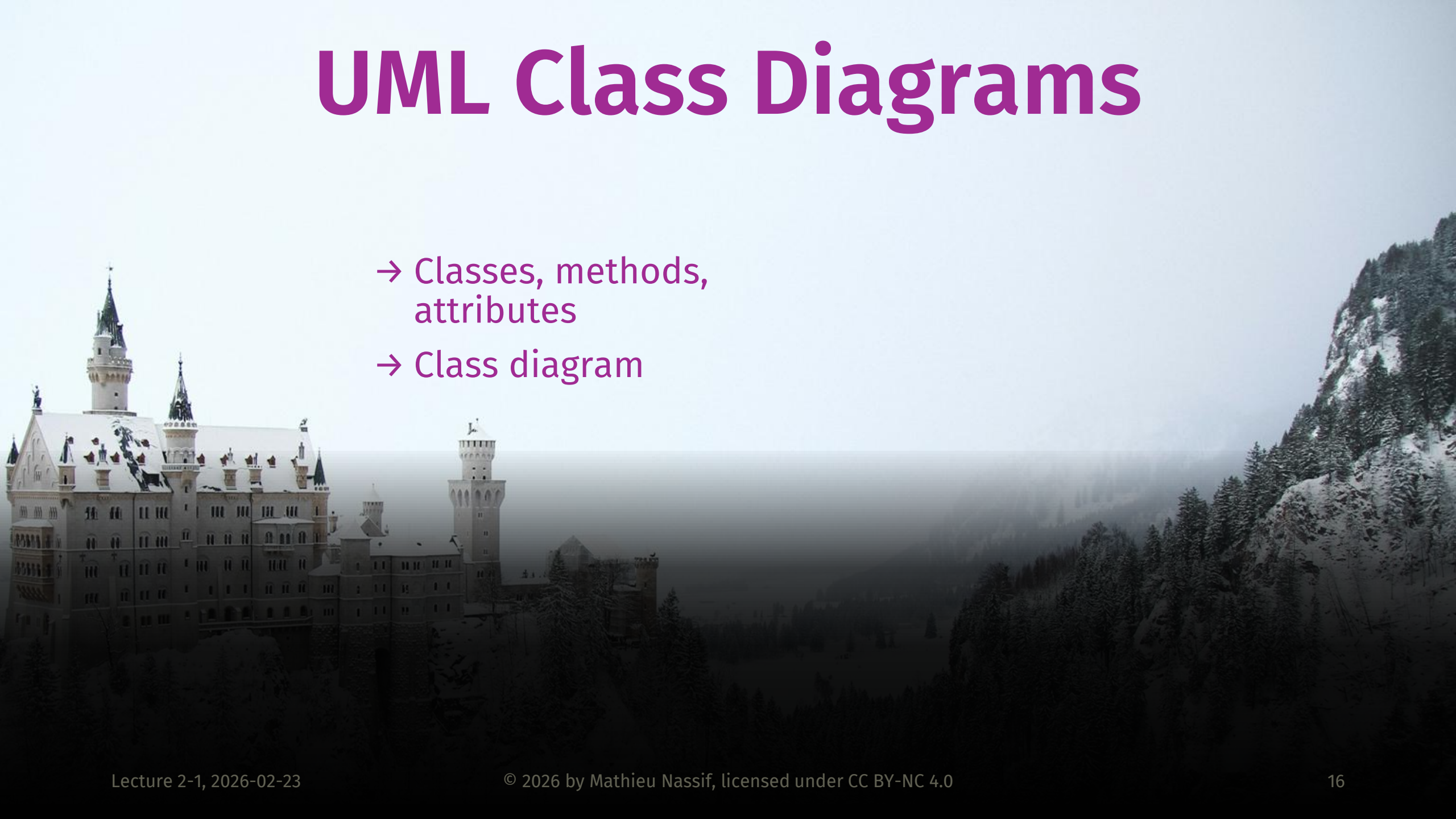

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“Write once, run everywhere”



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UML Class Diagrams

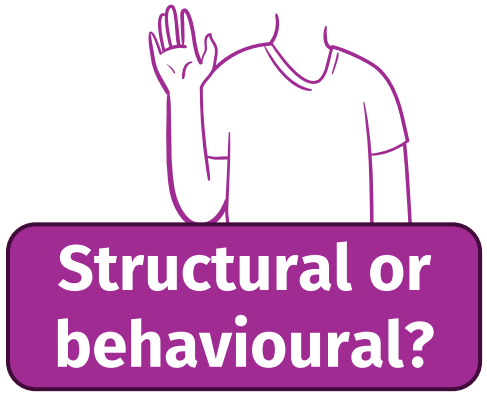


- → Classes, methods, attributes
- → Class diagram

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UML Class Diagram

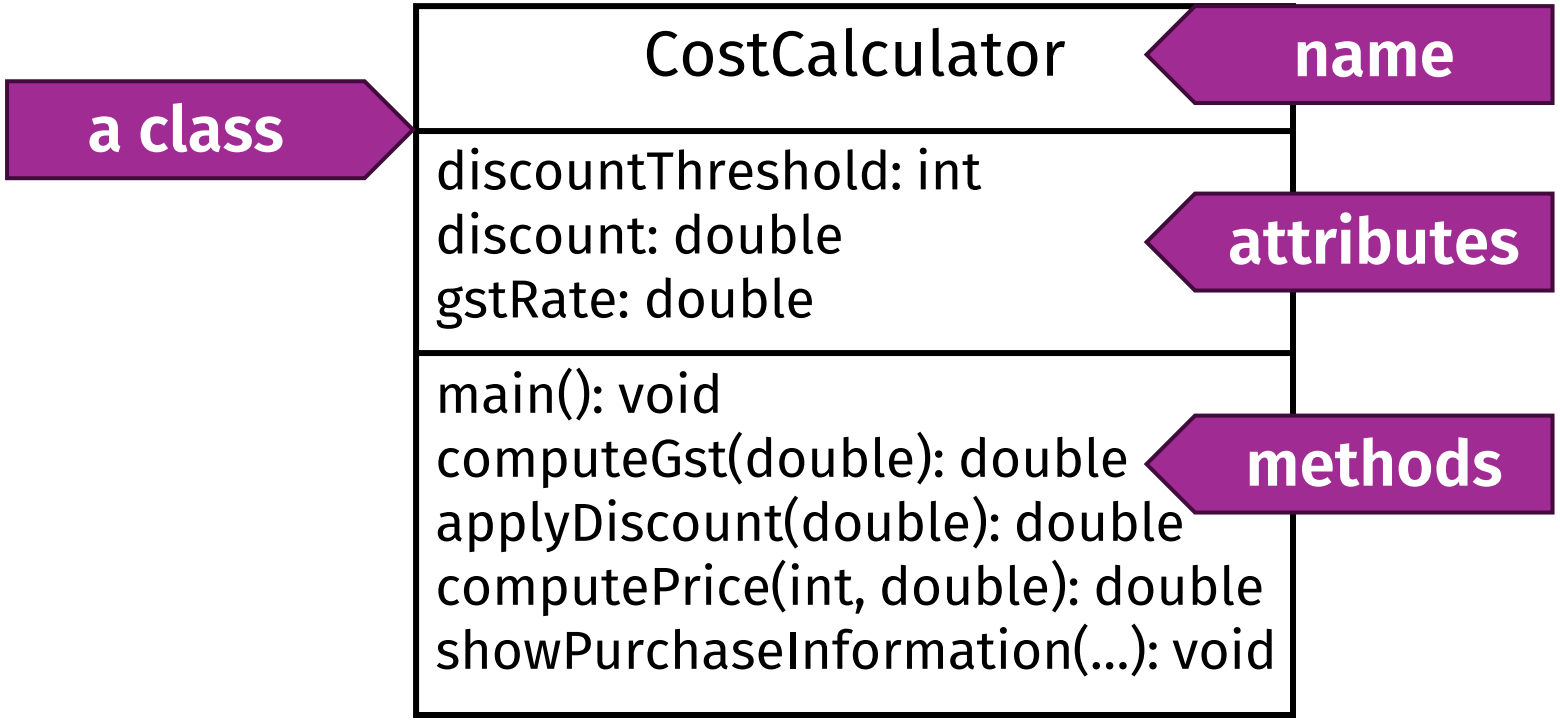
Models **classes** and their **dependencies**



- ✓ What are the abstractions of my program?
- ✓ Where is the data stored?
- ✓ Which part of the code provide which functionality?
- ✓ How do different classes interact with each other?

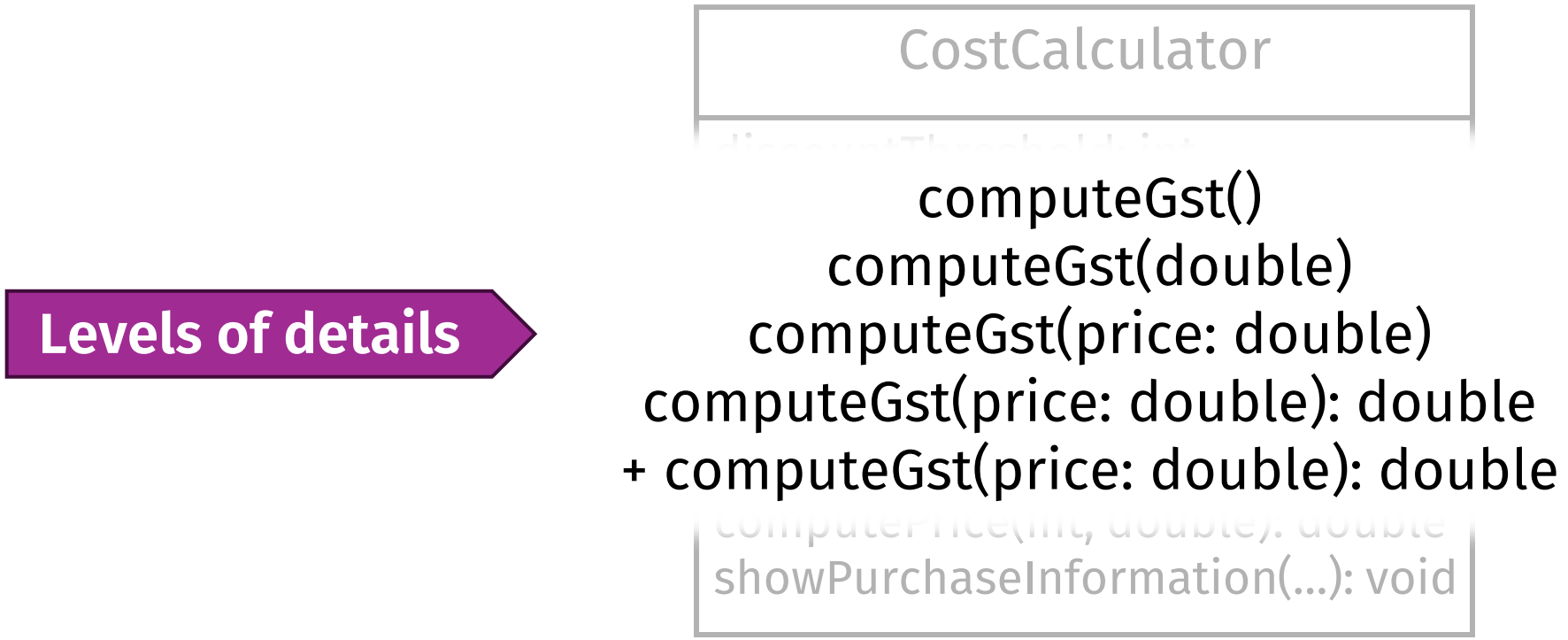
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UML Class Diagram



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UML Class Diagram



Levels of details

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UML Class Diagram



CostCalculator
discountThreshold: int discount: double gstRate: double
main(): void computeGst(double): double applyDiscount(double): double computePrice(int, double): double showPurchaseInformation(...): void

It's a model!

Include as much as needed to convey the important design information

AND be readable

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Exercise: draw the UML diagram

```java
package lecture2_1;

public class UmlClassDiagram {

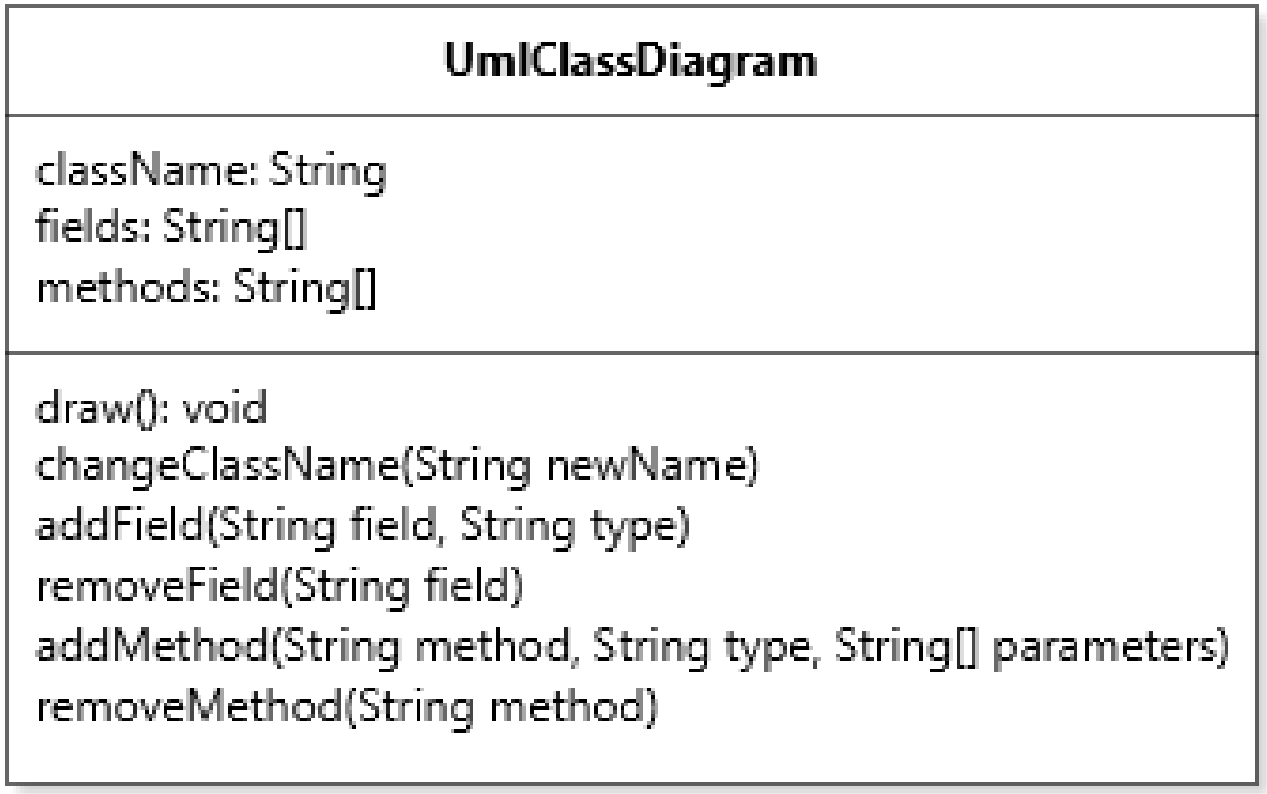
    String className = "Class";
    String[] fields = {};
    String[] methods = {};

    void draw() { /* ... */ }

    void changeClassName(String newName) { /* ... */ }
    void addField(String field, String type) { /* ... */ }
    void removeField(String field) { /* ... */ }
    void addMethod(String method, String type, String[] parameters) { /* ... */ }
    void removeMethod(String method) { /* ... */ }
}
```


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Exercise: draw the UML diagram



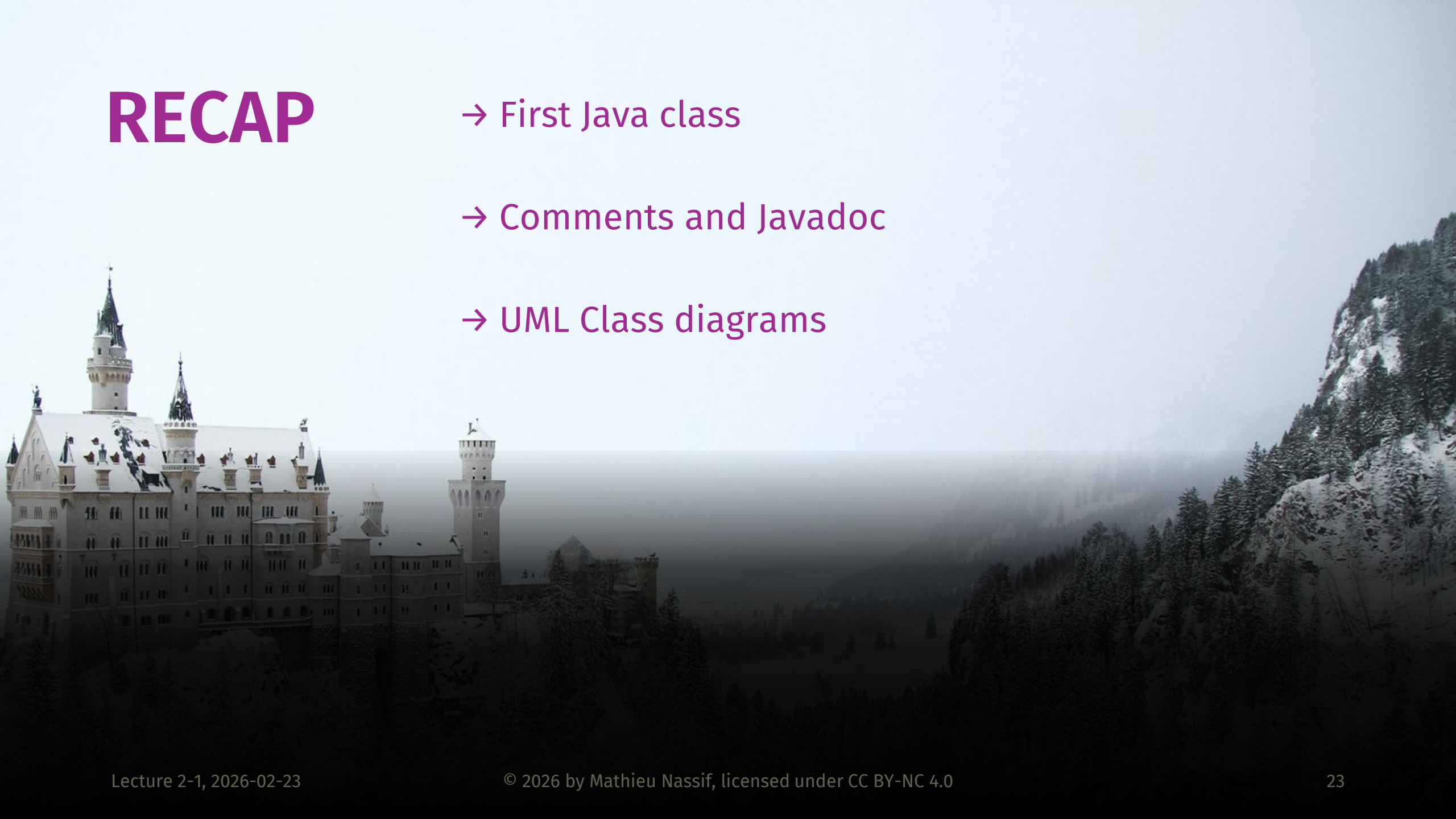
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RECAP

→ First Java class

→ Comments and Javadoc

→ UML Class diagrams



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